

Integrated Municipal Information System (IMIS)

MOBILE APP: SUPERVISOR CONTAINMENTASSESSMENT AND EMPTYING SERVICE DETAILS (SCA+ESD)

User Manual



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Solution(P) Ltd.

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Preface

This material will help Conservancy Supervisor from Conservation Section of Pourashava or Outsourced Agency Staff on the functionalities of Supervisor Containment Assessment (SCA) section.

This material will also help Vacutug Operators from Conservation Section of Pourashava or Outsourced Agency Staff on the functionalities and use of Emptying Service Details (ESD) section.

This material has been prepared to provide guidance to perform collection of containment assessment details and to perform details related to actual emptying operation.

Step by step procedure of all implemented facilities within the App is described with details.

- ☐ App Introduction
- ☐ Installation
- ☐ Login
- ☐ Dashboard
- ☐ Supervisor Containment Assessment
- ☐ Emptying Service Detail

Details related to actual emptying operation can also be entered in the system through Application section of Web IMIS as well.

1 App Introduction

1.1 Purpose

Supervisor Containment Assessment and Emptying Service Details for IMIS (SCA+ESD-IMIS) mobile app is built for performing initial assessment of containments that have applied for emptying request before actual emptying operation and finally for filling details related to actual emptying operation.

1.2 Sections

The SCA and ESD mobile app has two sections

1. Supervisor Containment Assessment (SCA)
2. Emptying Service Details (ESD)
3. Logout

1.3 Target User of app

Target User of the **SCA** section are:

- Conservancy Supervisor from Conservation Section of Jhenaidah Pourashava
- Outsourced Agency Staff.

Target User of the **ESD** section are:

- Vacutug Operators from Conservation Section of Jhenaidah Pourashava
- Outsourced Agency Staff.

2 Installation

An .apk file will be provided which can be easily downloaded and installed in your smartphones.

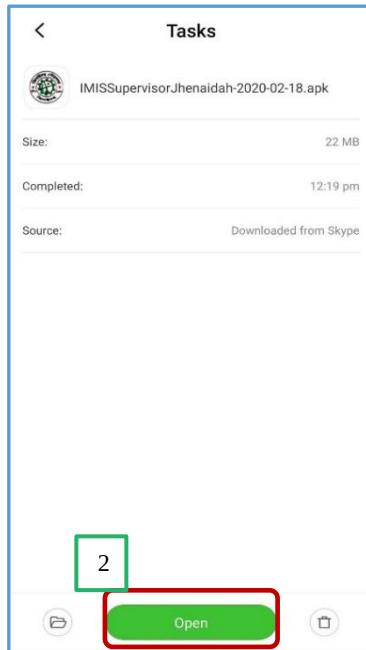
2.1 Prerequisite

Minimum Android 5.0

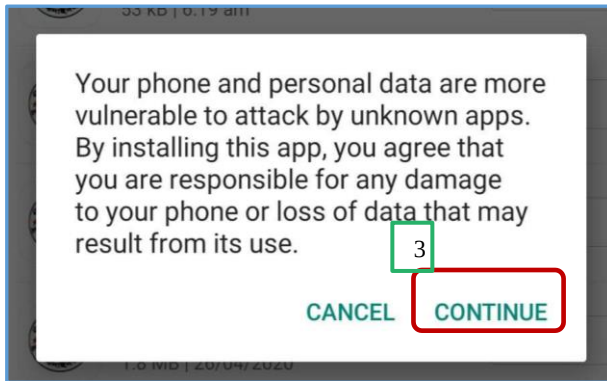
2.2 Installation Steps

Step 1: Download the .apk file in your smartphone. Then go to 'Download' section through the 'File Manager' of your smartphone.

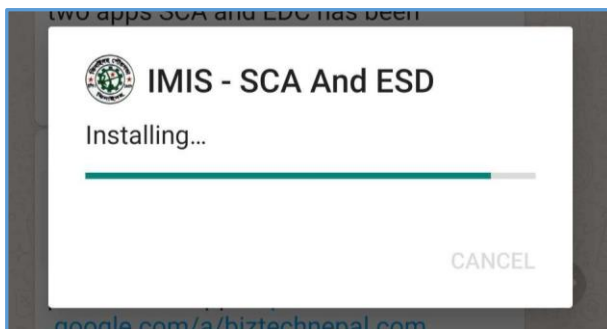
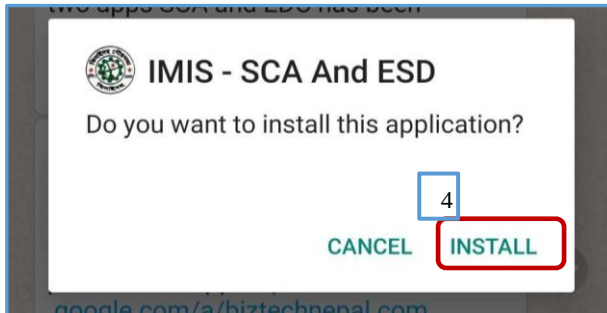
Step 2: Locate the .apk file and tap on it, and then Tap 'Open' button.



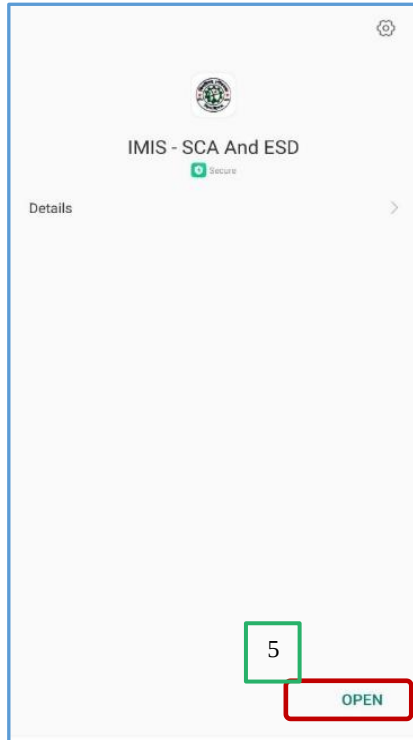
Step 3: Tap on 'Continue' button.



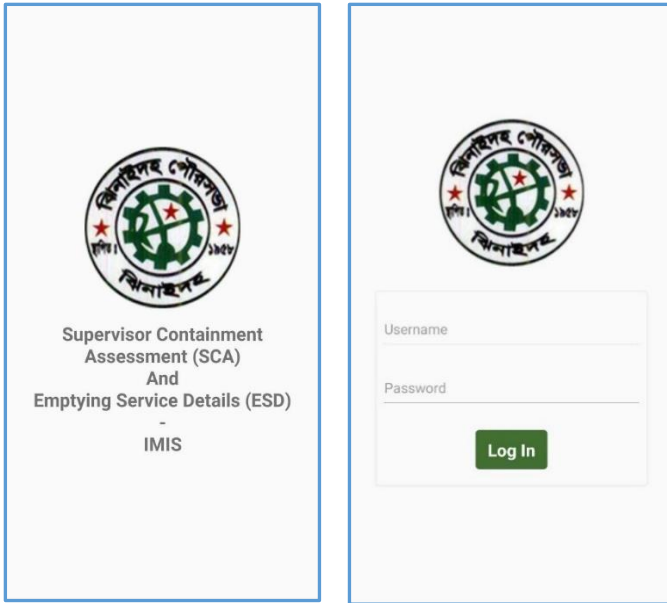
Step 4: Tap on 'Install' button.



Step 5: Tap on ‘Open’ button.



The app will be installed and a login screen will appear.



If the app is installed previously in the smart phone, then find the app and open the app to go to the login screen.



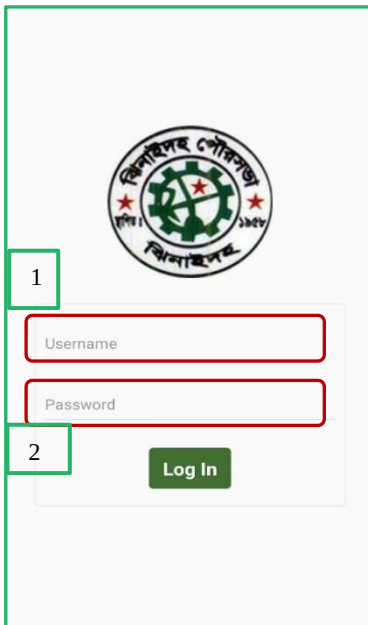
3 Login

The users/ supervisors/ emptying professionals are signed up through IMIS website with an email and password by the Admin or Help desk.

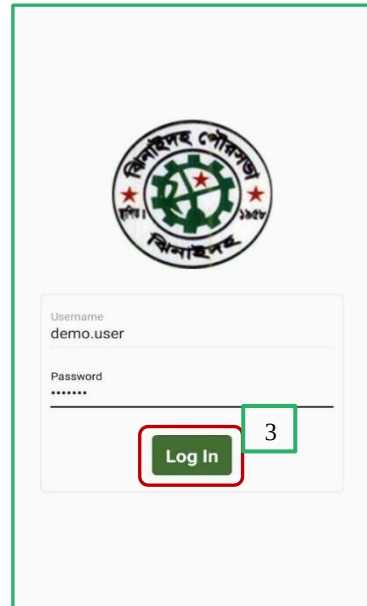
On opening the app for the first time, the user is asked for their login credentials.

The login credentials are saved in the app unless the user logout of the app or sign in from another device.

1. Enter username in the ‘Username’ textbox.
2. Enter password in ‘Password’ textbox.
3. Tap on ‘Login’ button.



This screenshot shows the login interface with a circular logo at the top. Below the logo are two text input fields: 'Username' and 'Password'. A green box labeled '1' highlights the 'Username' field, and another green box labeled '2' highlights the 'Password' field. At the bottom center is a green 'Log In' button.

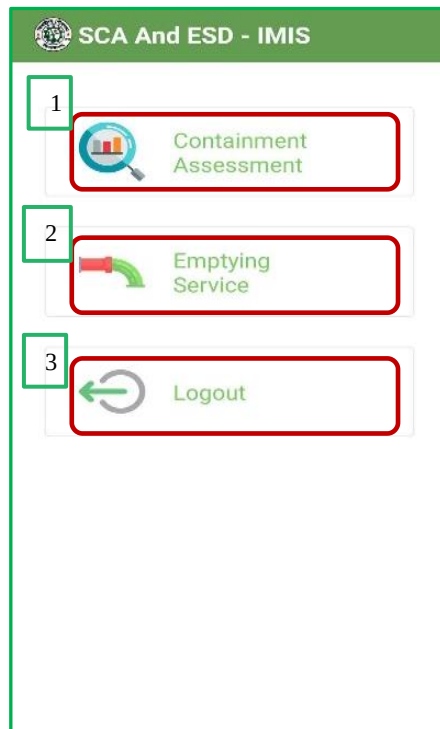


This screenshot shows the login interface with the same circular logo. The 'Username' field contains the text 'demo.user' and the 'Password' field contains a series of asterisks. A green box labeled '3' highlights the green 'Log In' button at the bottom.

4 Dashboard

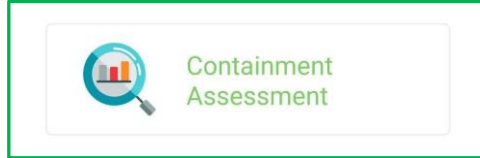
After successful login, a dashboard page will appear. The user has to select:

1. 'Containment Assessment' button to enter details related to containment assessment process. This is for the supervisor's use.
2. 'Emptying Service' button to enter emptying operation details. The Vacutug operators will select this section after emptying to fill up emptying details.
3. 'Logout' button to log out from the app after data input.



5 Supervisor Containment Assessment

This section is for the supervisors to enter assessment related information.



5.1 Containment Assessment

The user needs to fill up a form to after selecting ‘Containment Assessment’ button.

Fill the form and finally press ‘Submit’ button to upload the details to Web IMIS.

← Assessment Form	
Application Id *	Select Application ID
Service Provider *	Select Service Provider
Distance From Road (m) *	Enter Distance form road
Road Width (m) *	Enter width of road
Assessed Sluge (m3) *	Enter Assessed Sludge
Tank Length (m) *	Enter Tank Length
Tank Width (m) *	Enter Tank Width
Tank Depth (m) *	Enter Tank Depth
☒ Vacutug Accessibility?	
Width of Recommended Vacutug (ft) *	Select width of Recommended Vacutag
Size of Recommended Vacutug *	Select size of Recommended Vacutag
Required Trips *	Enter Number of trips Required
Comments *	Enter Comments if any
Proposed Emptied Date *	Select Proposed Emptying date from datepicker
Estimated Cost *	Enter Estimated Cost
SUBMIT	

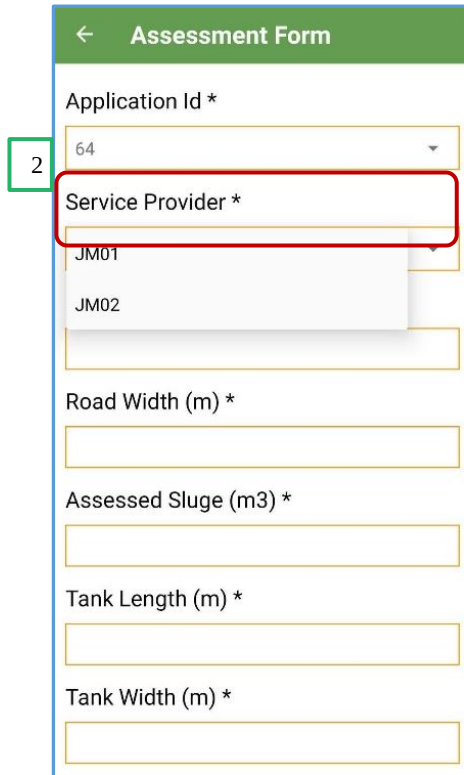
5.2 Update Information

Step 1: Select ‘Application ID’ from the dropdown menu.

The screenshot shows the 'Assessment Form' interface. A green header bar at the top contains a back arrow and the text 'Assessment Form'. Below this, a form field labeled 'Application Id *' is highlighted with a red border. To the left of this field, a green box with the number '1' indicates the first step. The dropdown menu is open, showing a list of application IDs: 24, 40, 43, 44, 56, 57, 58, 59, 61, and 63. The text 'Select Application Id' is displayed at the top of the list.

‘Application ID’ is the uniquely generated ID when an emptying application request is received beforehand. Only Application ID associated with containments which is yet to be assessed by supervisor will be shown in the dropdown list.

Step 2: Select ‘Service provider’ from the dropdown menu.



The screenshot displays the 'Assessment Form' interface. At the top, there is a green header bar with a back arrow and the title 'Assessment Form'. Below the header, the form contains several fields. The 'Application Id *' field is a dropdown menu currently showing '64'. To the left of this field, a green box with the number '2' is present. Below the 'Application Id' field is the 'Service Provider *' field, which is highlighted with a red rectangular border. This field is also a dropdown menu, and its list is open, showing two options: 'JM01' and 'JM02'. Below the 'Service Provider' field are four more text input fields: 'Road Width (m) *', 'Assessed Sluge (m3) *', 'Tank Length (m) *', and 'Tank Width (m) *'.

‘Service Provider’ is the name of the party providing with assessment service. The enlisted parties will be shown in the dropdown list from where it can be selected.

Step 3: ‘Distance from Road (m)’ is the distance of the building’s containment system from the main road in meter unit.

Step 4: ‘Road Width (m)’ is the width of the nearest road to the containment in meter unit.

Step 5: ‘Estimated Assessed Sludge (m3)’ is the estimation of sludge volume in cubic meter unit. It will be estimated by the supervisor through visual inspection.

← **Assessment Form**

Application Id *

Select Applicaiton Id ▼

Service Provider *

Select Service Provider ▼

3 Distance From Road (m) *

4 Road Width (m) *

5 Estimated Assessed Sluge (m3) *

☐ Vacutug Accessibility?

Required Trips *

Step 6: Depending upon containment type (septic tank/pit) associated with selected Application ID, fields related containment size will dynamically change.

If not, enter the dimensions manually.

← Assessment Form

Application Id *

64

Service Provider *

JM01

Distance From Road (m) *

Road Width (m) *

Assessed Sludge (m3) *

6

Tank Length (m) *

Tank Width (m) *

For **Pit type** containment associated with the application enter pit dimension. For **Septic tank type** containment associated with the application enter tank dimensions

Number Of Pit *	Septic Tank Length (m) *
Pit Diameter (m) *	Septic Tank Width (m) *
Pit Depth (m) *	Tank Depth (m) *

Step 7: Checking the ‘Vacutug Accessibility’ button will help to determine the size of the Vacutug which can go through the roads and access the containment system for emptying.

Step 8: A tap on ‘Width of Recommended Vacutug (ft)’ will show a dropdown menu from where width of the Vacutug can be selected in foot units.

Step 9: The ‘Size of Recommended Vacutug’ is the volume capacity in Liters. It will appear with respect to the selected width.

← Assessment Form

Pit Depth (m) *

7 ☐ Vacutug Accessibility?

8 Width of Recommended Vacutug (ft) *

Select Width

9 Size of Recommended Vacutug *

Select Size

Required Trips *

Comments *

Add your comments here

Step 10: ‘Required trips’ is the number of trips need to be done by the Vacutug for emptying the containment.

Step 11: In the ‘Comments’ text box, Supervisor can leave any comments on the containment system.

Step 12: ‘Proposed Emptying Date’ needs to be filled up by consulting with the customer.

Step 13: ‘Estimated Cost’ needs to be entered by calculating sludge volume and number of trips required. The cost will vary with different service providers.

← **Assessment Form**

Size of Recommended Vacutug *

Size of Recommended Vacutug is required.

10 Required Trips *

Required Trips is required.

11 Comments *

Add your comments here

Comments is required.

12 Proposed Emptied Date *

2020-04-30

13 Estimated Cost *

Estimated Cost is required.

SUBMIT

Step 14: Tap ‘Submit’ to upload information in Web IMIS.

← **Assessment Form**

Size of Recommended Vacutug *

Size of Recommended Vacutug is required.

Required Trips *

Required Trips is required.

Comments *

Add your comments here

Comments is required.

Proposed Emptied Date *

2020-04-30

Estimated Cost *

Estimated Cost is required.

14

SUBMIT

Step 15: Tap ‘OK’ in the ‘Data submitted successfully’ wizard.

Logout

Success

Data submitted successfully.

15

OK

5.3 Data Validation for Empty fields

If any required field is left empty, it will show up in red text and the assessment form cannot be submitted.

The screenshot shows a mobile application interface for an "Assessment Form". The form has a green header bar with a back arrow and the title "Assessment Form". Below the header, there are five input fields, each with a red asterisk indicating it is required. The first field is "Size of Recommended Vacutug", the second is "Required Trips", the third is "Comments", the fourth is "Proposed Emptied Date", and the fifth is "Estimated Cost". Each field is empty, and a red error message "X is required." is displayed below each field. The "Proposed Emptied Date" field shows the date "2020-04-30". At the bottom right of the form is a green "SUBMIT" button.

← Assessment Form

Size of Recommended Vacutug *

Size of Recommended Vacutug is required.

Required Trips *

Required Trips is required.

Comments *

Add your comments here

Comments is required.

Proposed Emptied Date *

2020-04-30

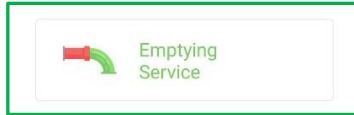
Estimated Cost *

Estimated Cost is required.

SUBMIT

6 Emptying Service Details

This section is for Vacutug operators to enter emptying service-related information into the IMIS database.



6.1 Emptying Service

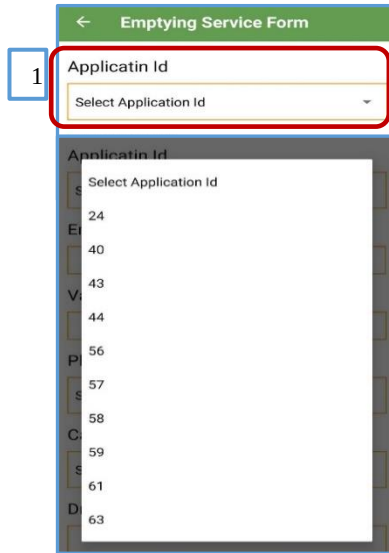
The emptying personnel need to fill up a form to after selecting 'Emptying Service' button.

Fill the form and finally press 'Confirm' and then 'Save' button to upload the details to Web IMIS.

← Emptying Service Form	
Application Id	
Select Application Id	Select Application ID
Emptied Date	
2020-04-12	Select Emptied Date from datepicker
Vacutug Number	
	Enter Vaccutag Number
Place of Disposal	
Select place of disposal	Select place of Disposal (Treatment Plant)
Capacity (ltr)	
Select Capacity	Select Capacity of Vaccutag
Driver	
	Enter name of driver
Emptier 1	
	Enter name of Emptier 1
Emptier 2	
	Enter name of Emptier 2
Start Time	
2:56 pm	Select Start Time
End Time	
3:56 pm	Select End Time
Time Required for Emptying Service After Arrival of Vacutug (in hr)	
	Enter time required
Required Trip for Total Emptying	
	Enter required trip
Receipt Number	
	Enter Receipt Number
Cost Paid by Containment Owner with Receipt	
	Enter Cost paid with Receipt
Cost Paid by Containment Owner without Receipt	
	Enter Cost Paid without Receipt
Cost of Disposal	
	Enter Cost of Disposal
Volume of Sludge (cum)	
	Enter Volume of Sludge
CONFIRM	

6.2 Update Information

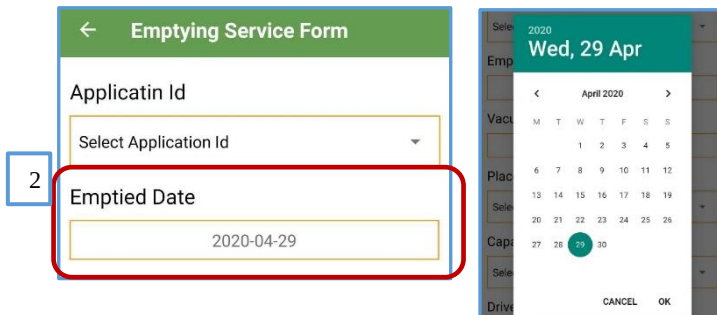
Step 1: ‘Application ID’ is the uniquely generated ID when an emptying application request is received beforehand.



The screenshot shows the 'Emptying Service Form' with a green header. Below the header, there is a section labeled 'Applicatin Id' (note the typo) which contains a dropdown menu with the text 'Select Application Id'. A red rectangular box highlights this dropdown menu. To the left of the box, there is a blue square containing the number '1'.

Only Application IDs associated with containments which is yet to be emptied will be shown in the dropdown list. User have to select the relevant ID and start entering details.

Step 2: ‘Emptied Date’ is the date on which the emptying service was provided and can be selected from a dropdown calendar.



The screenshot shows the 'Emptying Service Form' with a green header. Below the header, there is a section labeled 'Emptied Date' (note the typo) which contains a date field showing '2020-04-29'. A red rectangular box highlights this date field. To the left of the box, there is a blue square containing the number '2'. To the right of the main form, there is a calendar widget showing the date 'Wed, 29 Apr' for the year 2020. The calendar shows the month of April with days 1 through 30. The date 29 is highlighted in green.

Step 3: ‘Vacutug Number’ is the number of the Vacutug which is used for collecting sludge. Put the number in the text box.

← Emptying Service Form

Applicatin Id
Select Application Id ▼

Emptied Date
2020-04-29

3 Vacutug Number

Place of Disposal
Select place of disposal ▼

Step 4: ‘Place of Disposal’ is the name of the treatment plant, where the collected sludge will be taken for processing.

← Emptying Service Form

Applicatin Id
Select Application Id ▼

Emptied Date
2020-04-29

Vacutug Number

4 Place of Disposal
Select place of disposal ▼

Select place of disposal

- Treatment Plant 2
- Treatment Plant 4
- Treatment Plant 1
- Shailesh Shrestha
- TP1

It can be selected from a dropdown menu.

Step 5: ‘Capacity (ltr)’ is the volume capacity of the Vacutug in Liter unit. This has to be selected from a dropdown menu.

Step 6: In the ‘Driver’ option the name of the Vacutug driver needs to be filled in as text.

Step 7: In the ‘Emptier 1’ option the name of the emptier needs to be filled in as text.

Step 8: In the ‘Emptier 2’ option the name of the other emptier needs to be filled in as text.

← Emptying Service Form

5 Capacity (litr)
2000

6 Driver

7 Emptier 1

8 Emptier 2

Start Time
1:06 pm

End Time
1:06 pm

Time Required for Emptying Service
After Arrival of Vacutug (in hr)

Step 9: 'Start Time' and 'End time' needs to be selected from a dropdown clock.

← Emptying Service Form

9 Start Time

3:42 pm

End Time

3:42 pm

13:06

CANCEL OK

Step 10: 'Required Trip for Total Emptying' is the number of trips required for emptying the containment.

10 Required Trip for Total Emptying

Receipt Number

Cost Paid by Containment Owner with Receipt

Step 11: ‘Receipt Number’ is the number of receipts of the service provided.

Step 12: ‘Cost Paid by Containment Owner with Receipt’ is the amount of money paid by the owner with receipt.

Step 13: ‘Cost Paid by Containment Owner without Receipt’ is the amount of money paid by the owner without receipt.

← Emptying Service Form

Required trip for total Emptying

11 Receipt Number

12 Cost Paid by Containment Owner with Receipt

13 Cost Paid by Containment Owner without Receipt

Cost of Disposal

Volume of Sludge (cum)

CONFIRM

Step 14: ‘Cost of Disposal’ is the amount of money paid in the disposal zone.

Step 15: ‘Volume of Sludge (cum)’ is the volume of sludge emptied in cubic meter unit.

Step 16: After filling up all this information tap ‘Confirm’ button.

← Emptying Service Form

Required Trip for Total Emptying

Receipt Number

Cost Paid by Containment Owner with Receipt

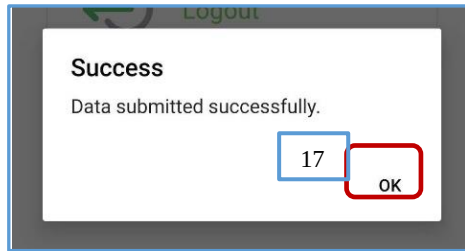
Cost Paid by Containment Owner without Receipt

14 Cost of Disposal

15 Volume of Sludge (cum)

16 CONFIRM

Step 17: A success confirmation dialog box will be displayed.
Press the 'OK' button.



IMIS FRAMEWORK

IMIS stands for **Integrated GIS-based Municipal Information System** developed for managing faecal sludge in Khulna City Corporation



KDA Detailed Area Plan (GIS-2014)

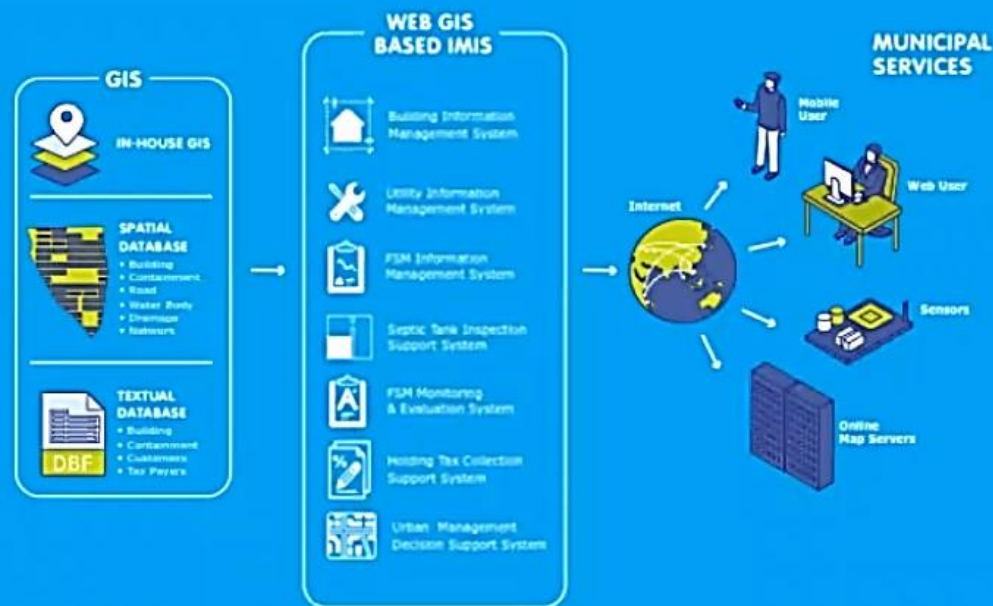
MAPS

- In house Data
- Spatial Database



Jhenaidah Municipality Information Framework (IMIS)

IMIS enables information based decision making and planning for **Municipal Services** in line with **SMART City** initiatives



IMIS is capable of providing city level data. For example:

